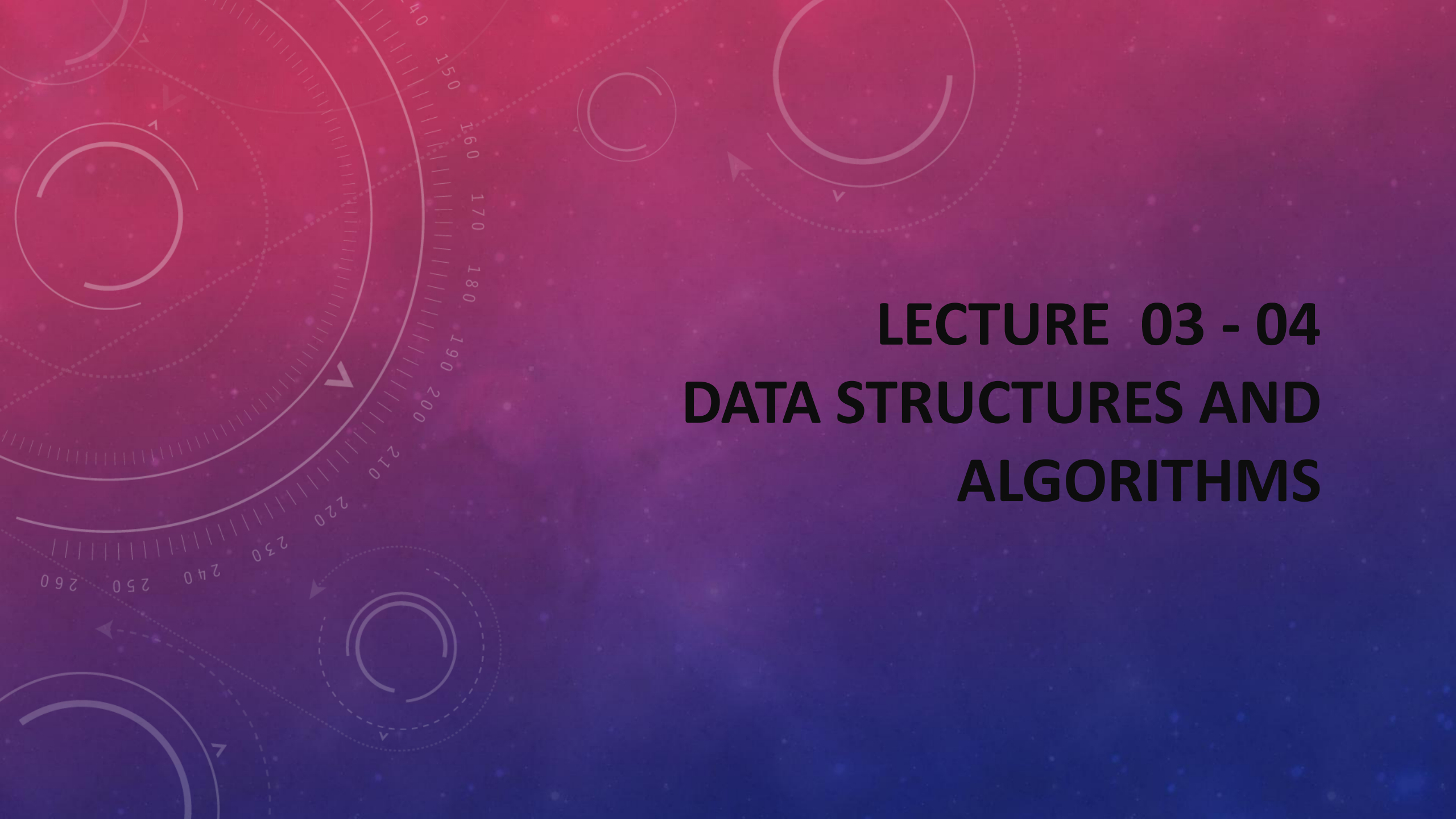


The background features a gradient from deep red at the top to dark blue at the bottom, speckled with white stars. Overlaid on the left side are several concentric circles and arcs in white and light blue. Some of these circles have tick marks and numbers, resembling a protractor or a circular scale. For example, one large circle has numbers 150, 160, 170, 180, 190, 200, 210, 220, 230, 240, 250, and 260. Other smaller circles and arcs are scattered across the left half of the image, some with arrows indicating direction.

LECTURE 03 - 04

DATA STRUCTURES AND

ALGORITHMS

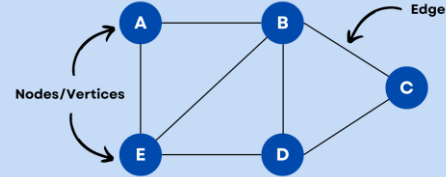
CONTENTS

DATA STRUCTURE OVERVIEW

DATA STRUCTURE VS DATA TYPE

VARIOUS DATA STRUCTURES

DATA STRUCTURE

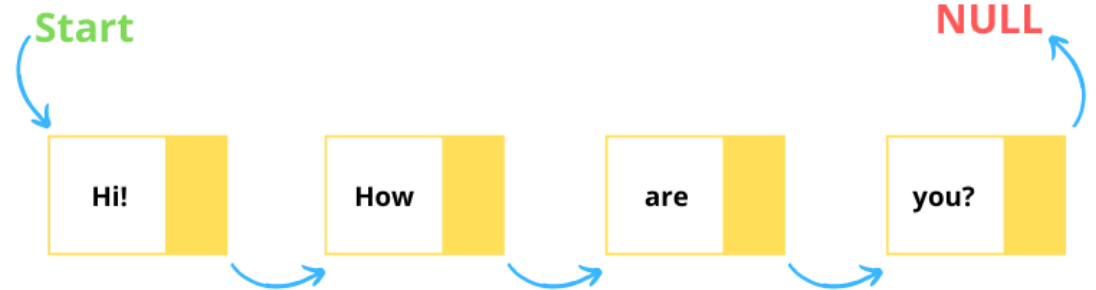


Address	11	12	13	14	15
Array	A	R	R	A	Y
Index	01	02	03	04	05

Lower Bound Upper Bound

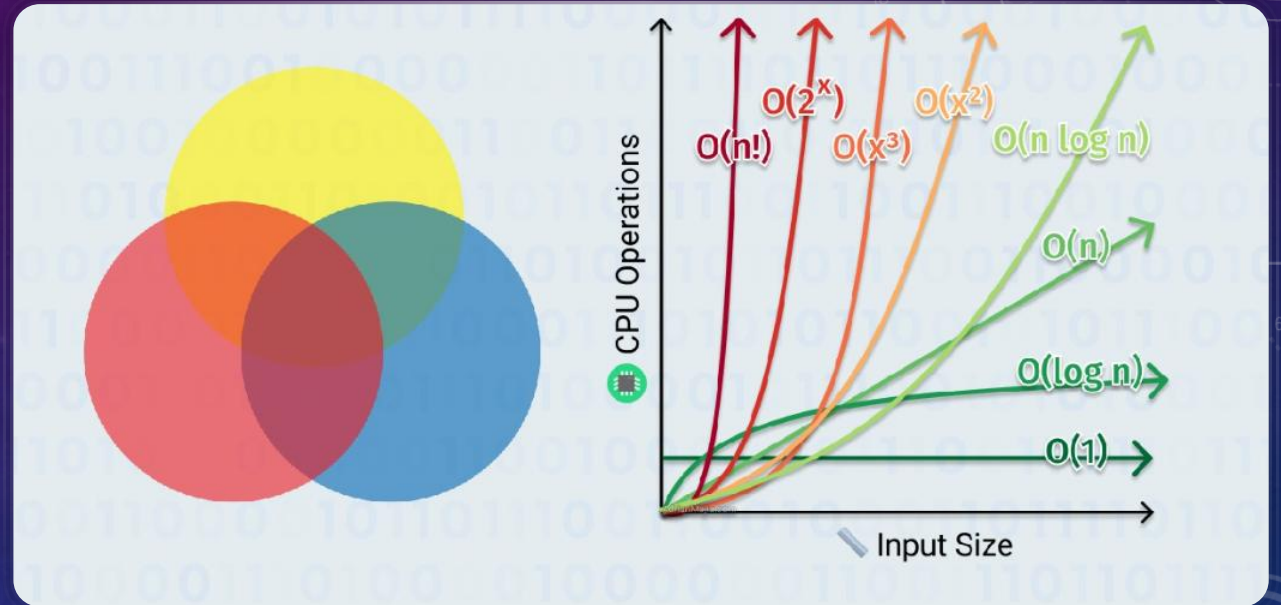
Data structure is a way of collecting and organizing data in such a way that we can perform operations on these data in effective way.

Linked list



DATA STRUCTURE

The three common goals of implementing a particular data structure are correctness, time complexity, and space complexity

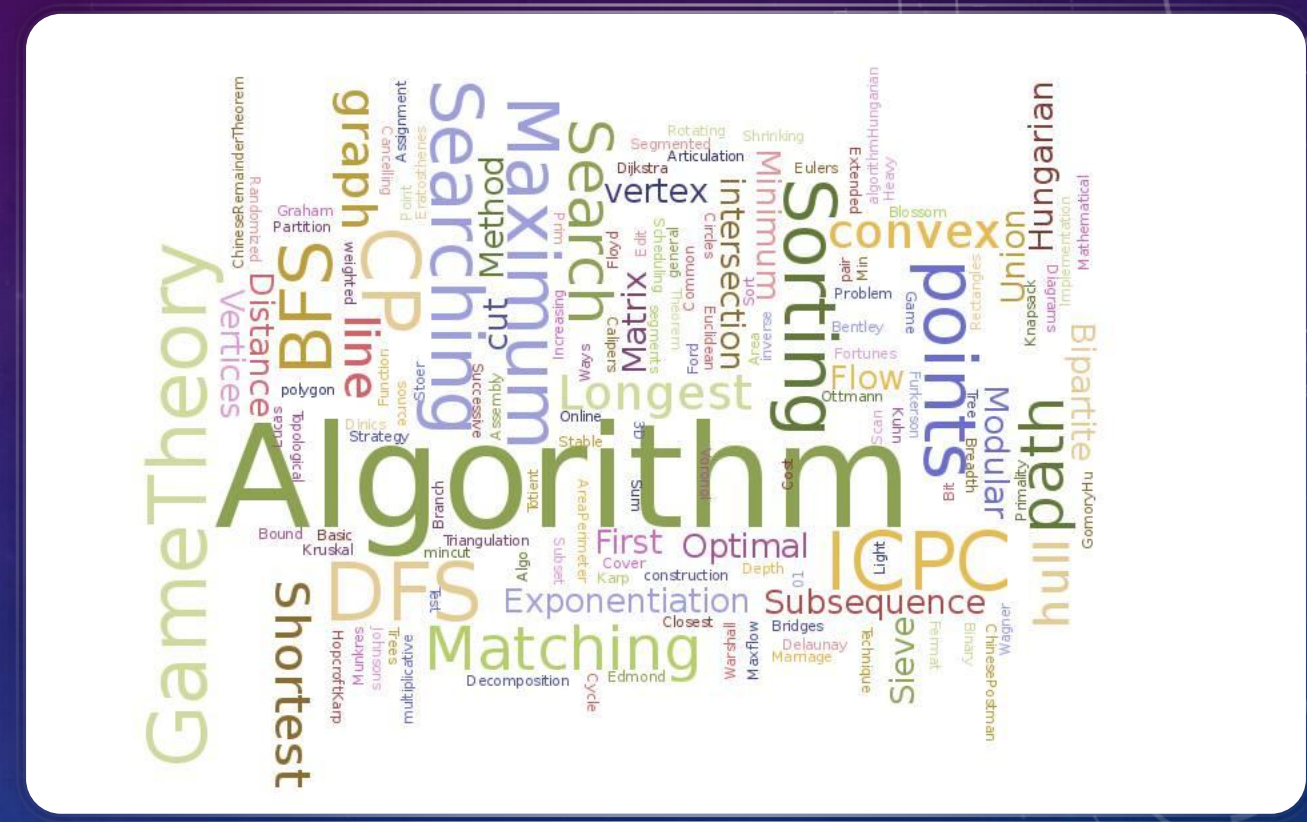


DATA STRUCTURE

Data structures came as a rescue to solve searching required information in exponentially increasing data, processor efficiency, and server capability.

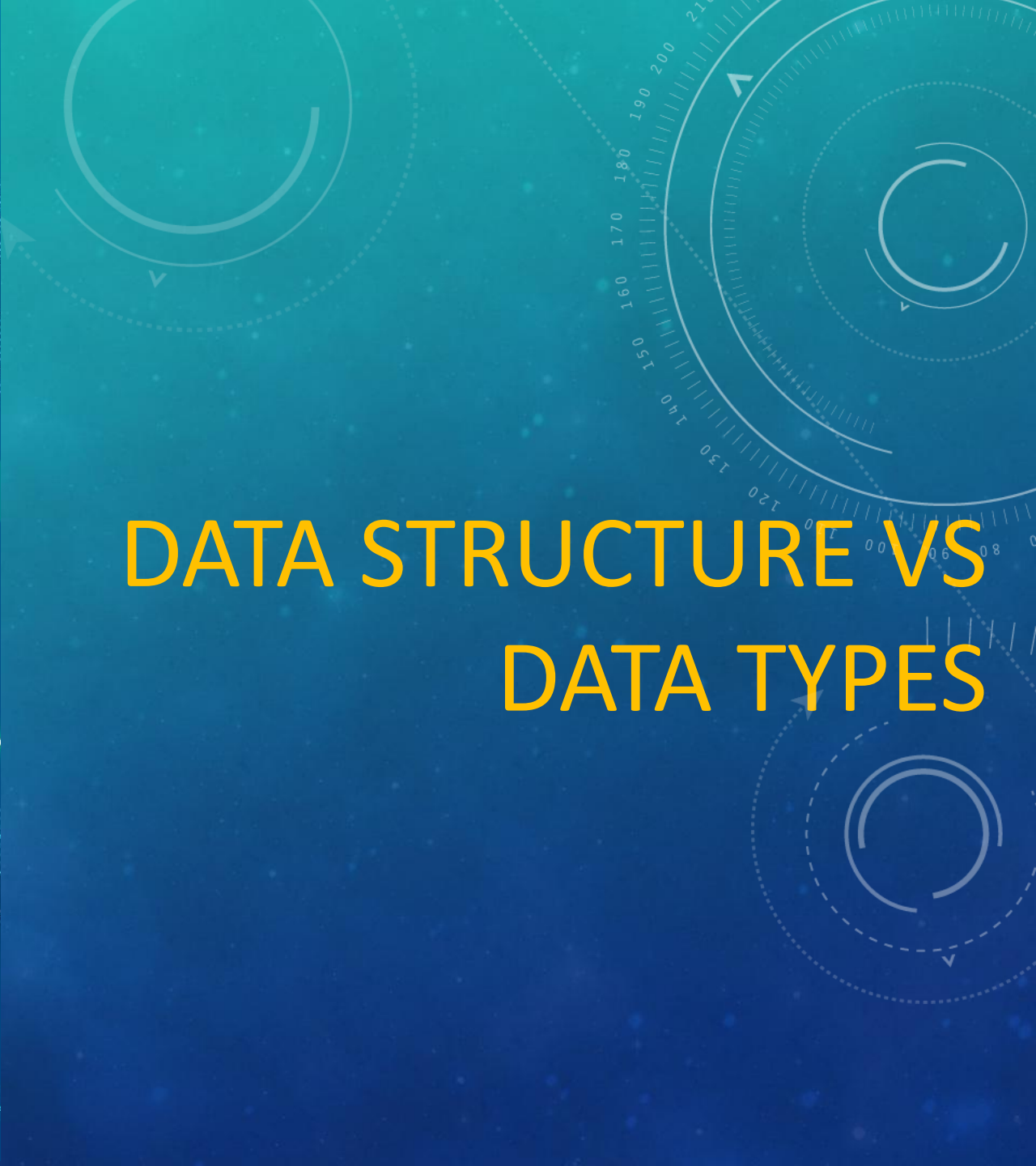


DATA STRUCTURE



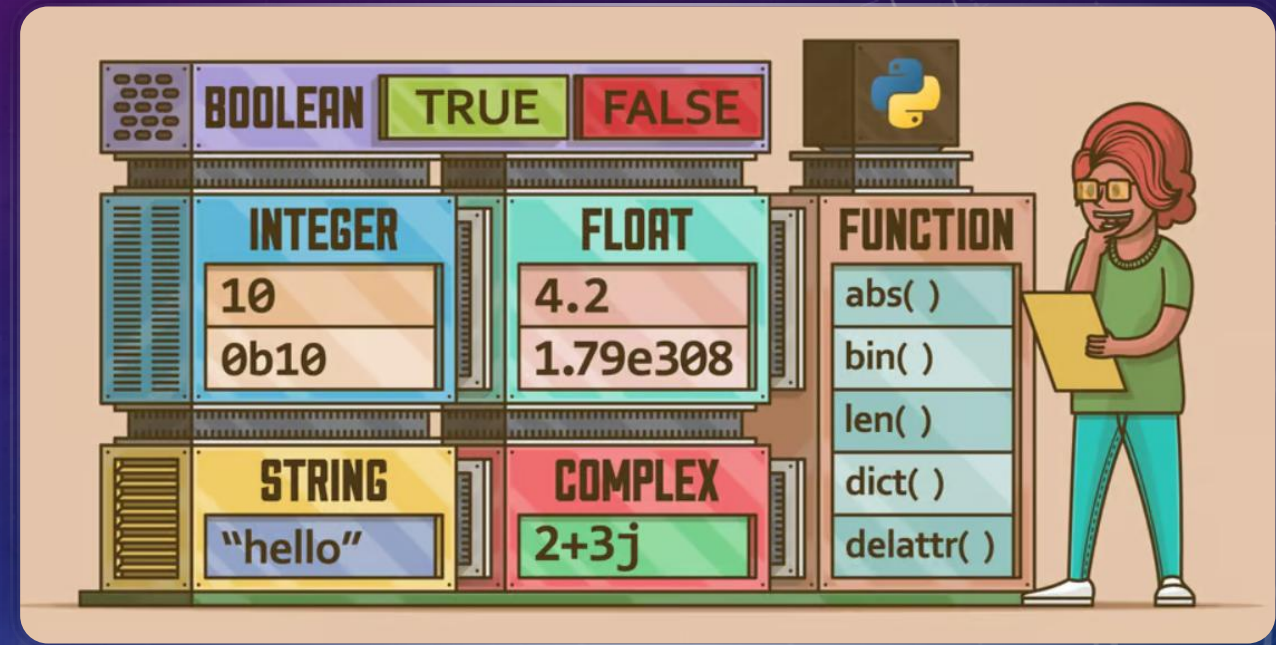


DATA STRUCTURE VS DATA TYPES



DATA TYPE

The data type is the form of a variable to which a value can be assigned. It defines that the variable will assign the values of the given data type only.



DATA TYPE(CONTD)

It can hold value but not data. Therefore, it is dateless.

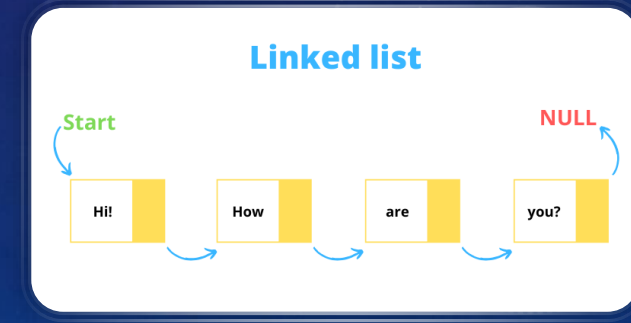
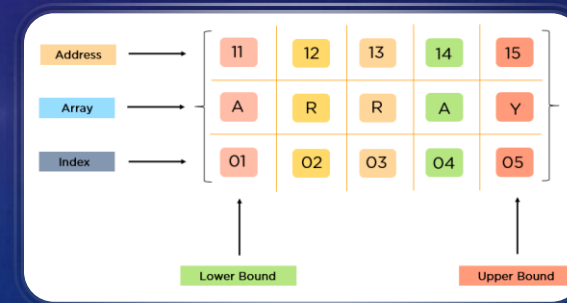
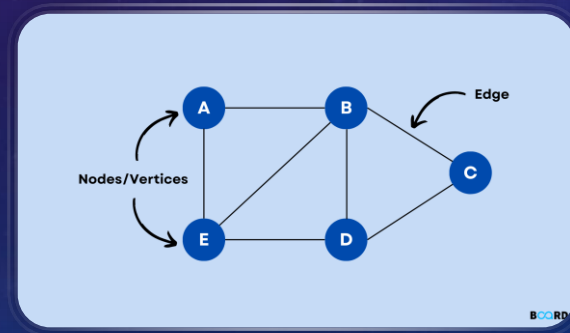
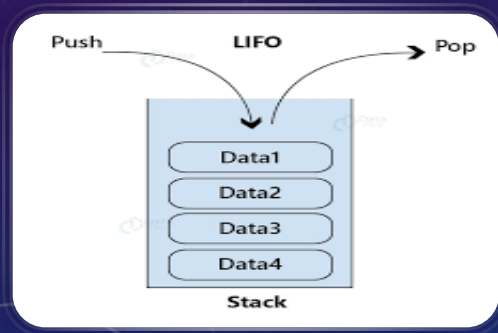
There is no time complexity in the case of data types.

In the case of data types, the value of data is not stored because it only represents the type of data that can be stored.

Data type examples are int, float, double, etc.

DATA STRUCTURE

Data structure is a collection of different kinds of data. That entire data can be represented using an object and can be used throughout the program.



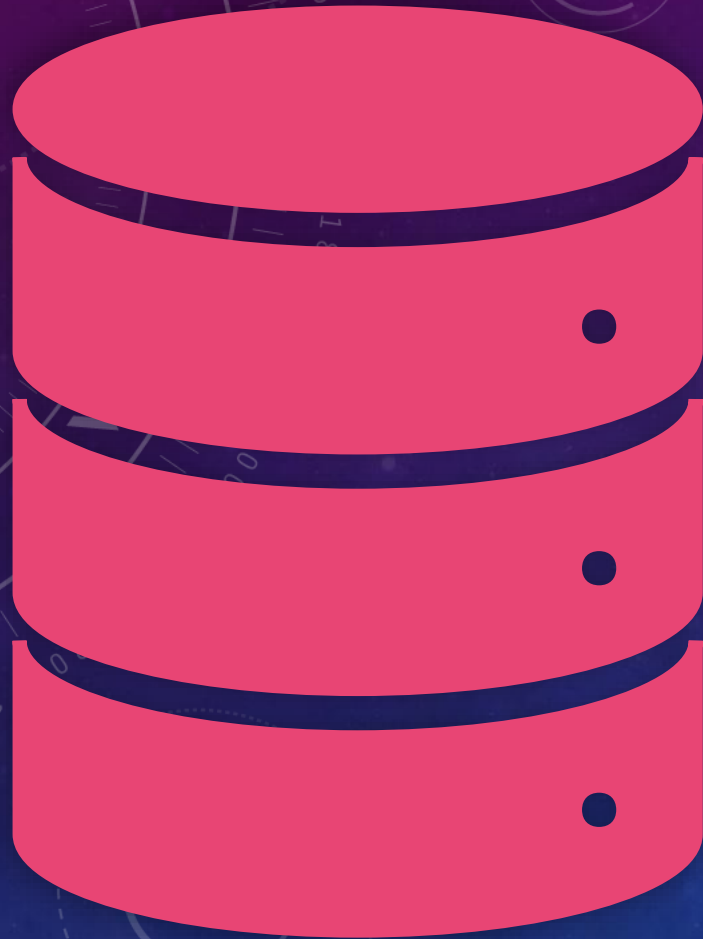
DATA STRUCTURE (CONTD)

It can hold multiple types of data within a single object.

In data structure objects, time complexity plays an important role.

While in the case of data structures, the data and its value acquire the space in the computer's main memory. Also, a data structure can hold different kinds and types of data within one single object.

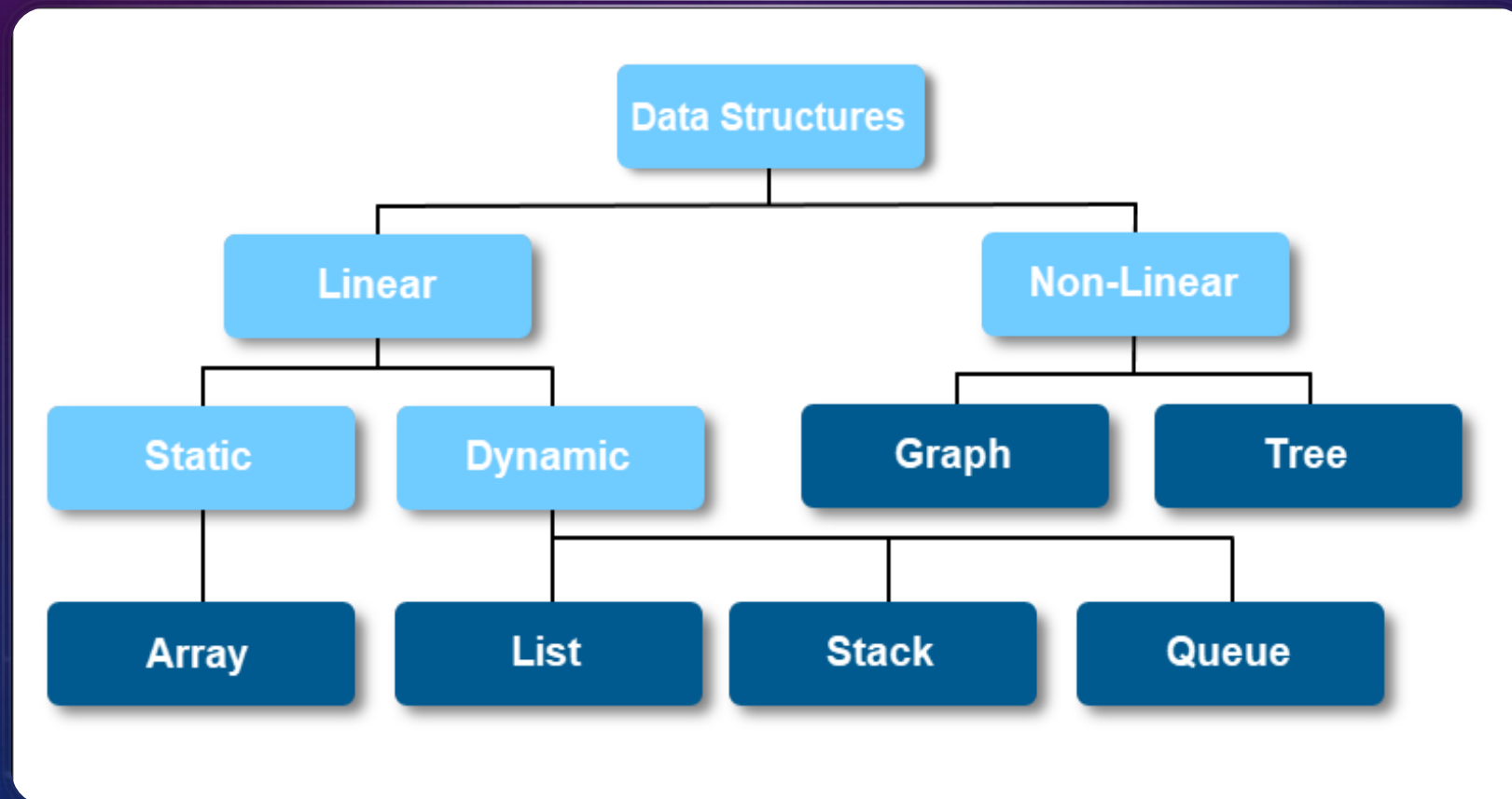
Data structure examples are stack, queue, tree, etc.



DATA STRUCTURE TYPES

DATA STRUCTURE TYPES

Basically, types of data structure have been categorized in two:



SO, HOW TO CHOOSE THE CORRECT DATA STRUCTURE

It depends on user requirements and effectiveness to choose a particular DS.

All DS provide different performance in different situations with different usage

The choice eventually impacts on performance efficiency, and resources such as memory and processor utilization.

For example: If the user wants to store data sequentially in the memory, the Array data structure will be effective.



DATA STRUCTURE TYPES

LINEAR DATA
STRUCTURE

LINEAR DATA STRUCTURE

In linear data structures, the elements are arranged in sequence one after the other.

Since elements are arranged in particular order, they are easy to implement.

However, when the complexity of the program increases, the linear data structures might not be the best choice because of operational complexities.

Array, Linked lists, Stack and Queue are popular linear DS

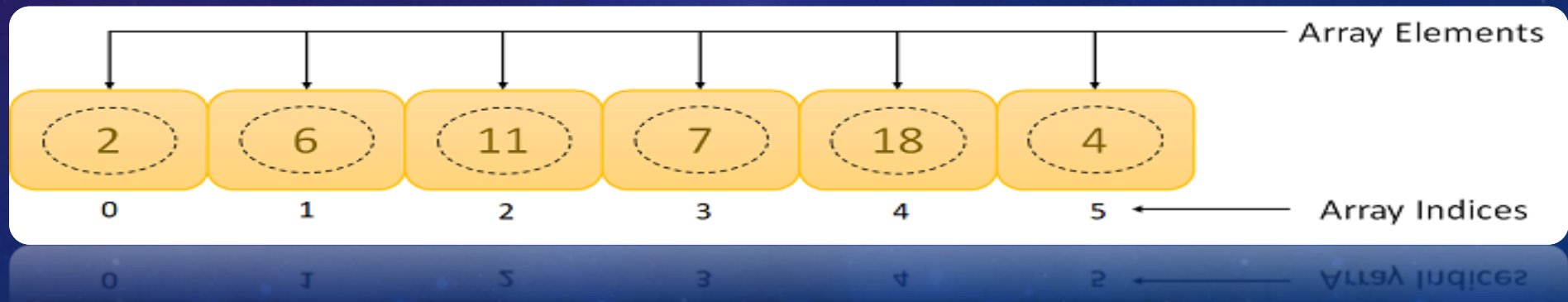
LINEAR DATA STRUCTURE - ARRAY

An Array is a data structure used to collect multiple data elements of the same data type into one variable. The data elements of the array share the same variable name

Instead of storing multiple values of the same data types in separate variable names, we could store all of them together into one variable.

An Array is a list of elements where each element has a unique place in the list , having different indices

Thus, the key feature of the arrays to understand is that the data is stored in contiguous memory locations, making it possible for the users to traverse through the data elements of the array using their respective indexes.



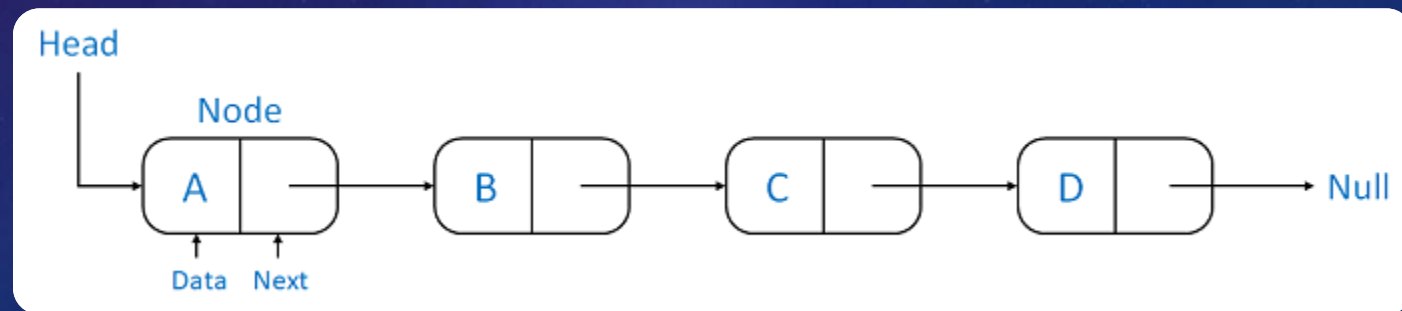
LINEAR DATA STRUCTURE – LINKED LIST

A Linked List is another example of a linear data structure used to store a collection of data elements dynamically.

Data elements in this data structure are represented by the Nodes, connected using links or pointers.

Each node contains two fields, the information field consists of the actual data, and the pointer field consists of the address of the subsequent nodes in the list.

The pointer of the last node of the linked list consists of a null pointer, as it points to nothing



LINEAR DATA STRUCTURE – STACK

In stack data structure, elements are stored in the **LIFO (Last In First Out)** principle. That is, the last element stored in a stack will be removed first.



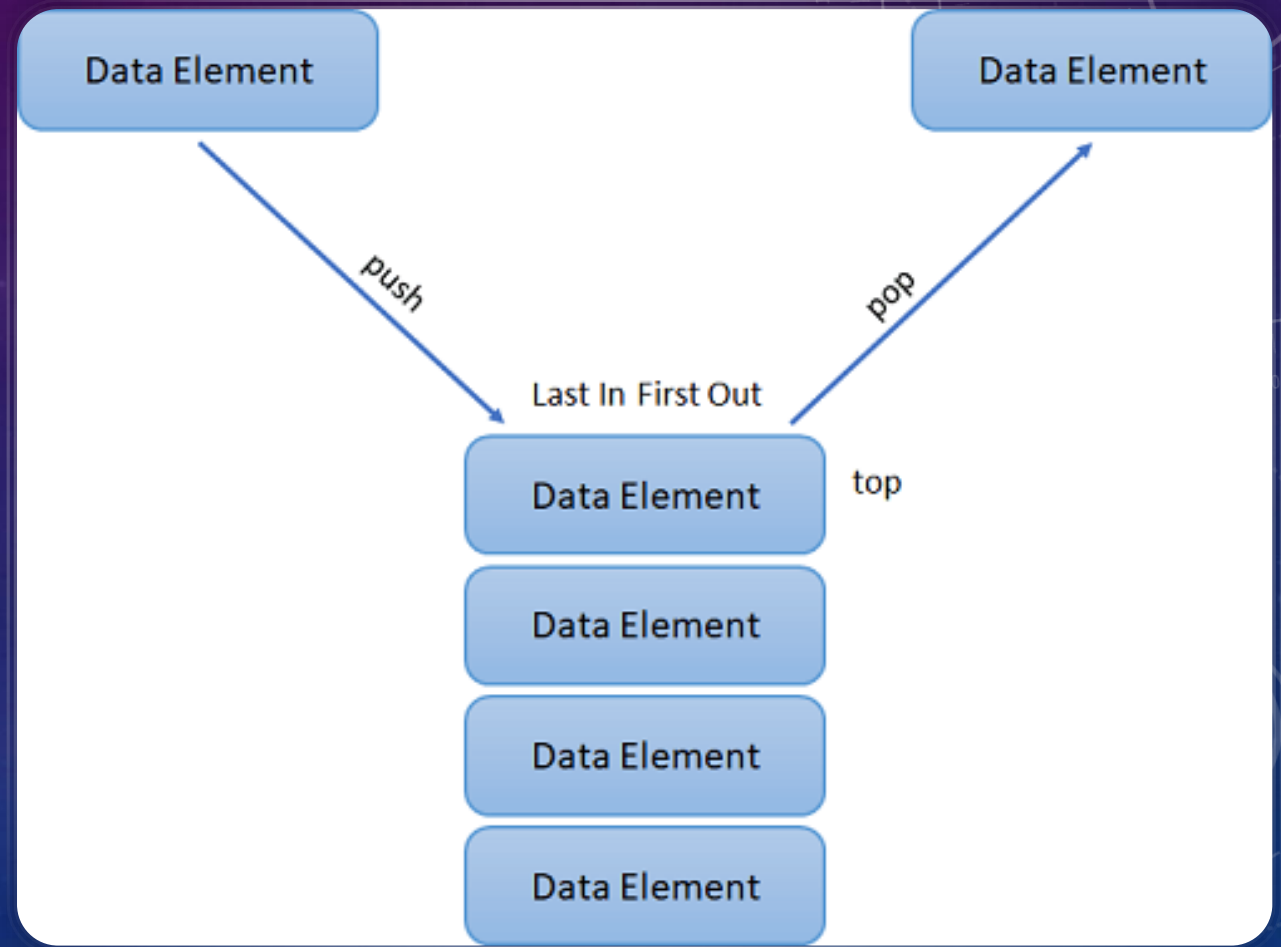
It works just like a pile of books where the last book kept on the pile will be removed first.

LINEAR DATA STRUCTURE – STACK

The primary operations in the Stack are as follows:

Push: Operation to insert a new element in the Stack is termed as Push Operation.

Pop: Operation to remove or delete elements from the Stack is termed as Pop Operation.



LINEAR DATA STRUCTURE – QUEUE

Queue data structure follows FIFO (First In, First Out) principle to manipulate the data elements. Implementation of Queues can be done using Arrays, Linked Lists, or Stacks.

The below image is a real-life illustration of a movie ticket counter that can help us understand the Queue where the customer who comes first is always served first.

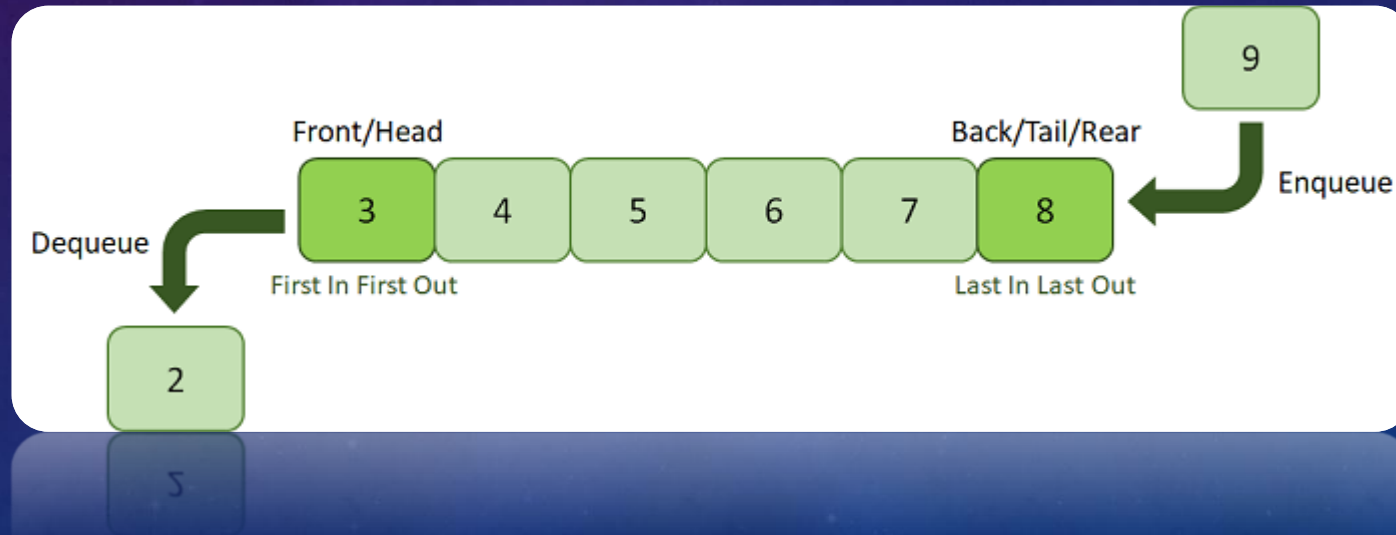


LINEAR DATA STRUCTURE – QUEUE

The following are the primary operations of the Queue:

Enqueue: The insertion or Addition of some data elements to the Queue is called Enqueue.

Dequeue: Deleting or removing data elements from the Queue is termed Dequeue.





DATA STRUCTURE TYPES

NON-LINEAR DATA STRUCTURE

NON-LINEAR DATA STRUCTURE

Unlike linear data structures, elements in non-linear data structures are not in any sequence.

Instead, they are arranged in a hierarchical manner where one element will be connected to one or more elements.

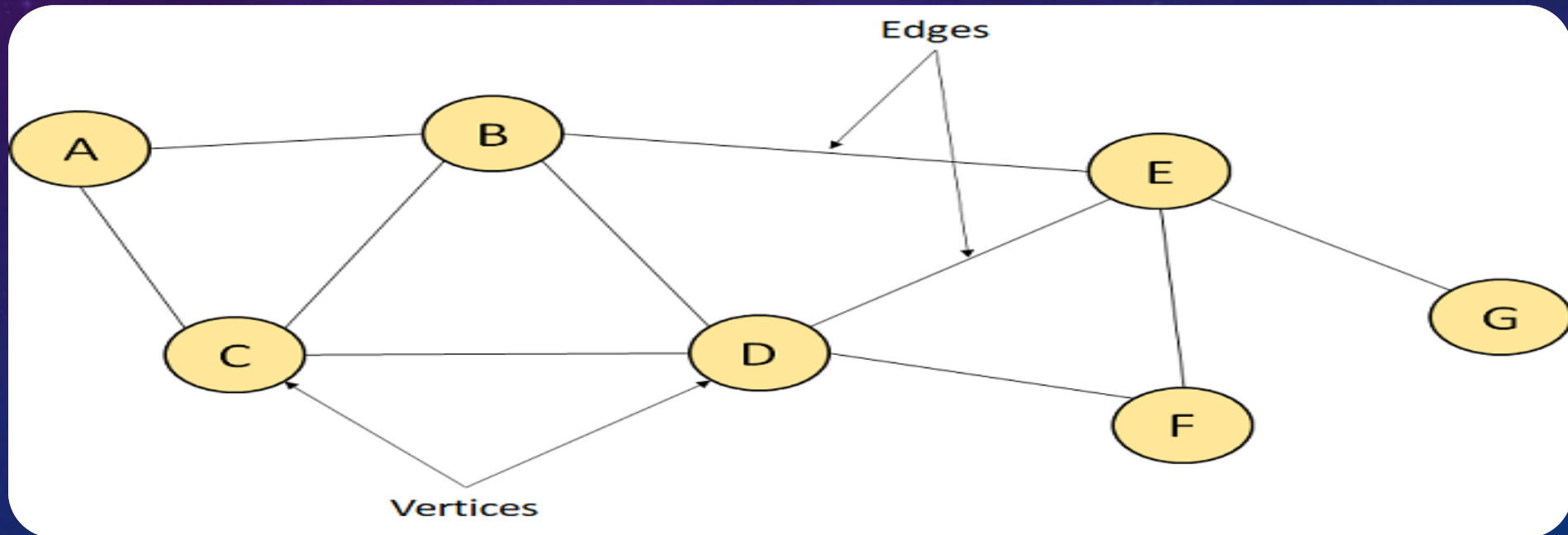
Graphs and Trees are popular non-linear DS

NON-LINEAR DATA STRUCTURE – GRAPHS

In Graph DS, the data elements are stored in such a way that it makes a graphical representation with respect to different relationship of elements.

There are nodes which stores data elements and edges which connect a node with adjacent one.

In graph data structure, each node is called vertex, and each vertex is connected to other vertices through edges.

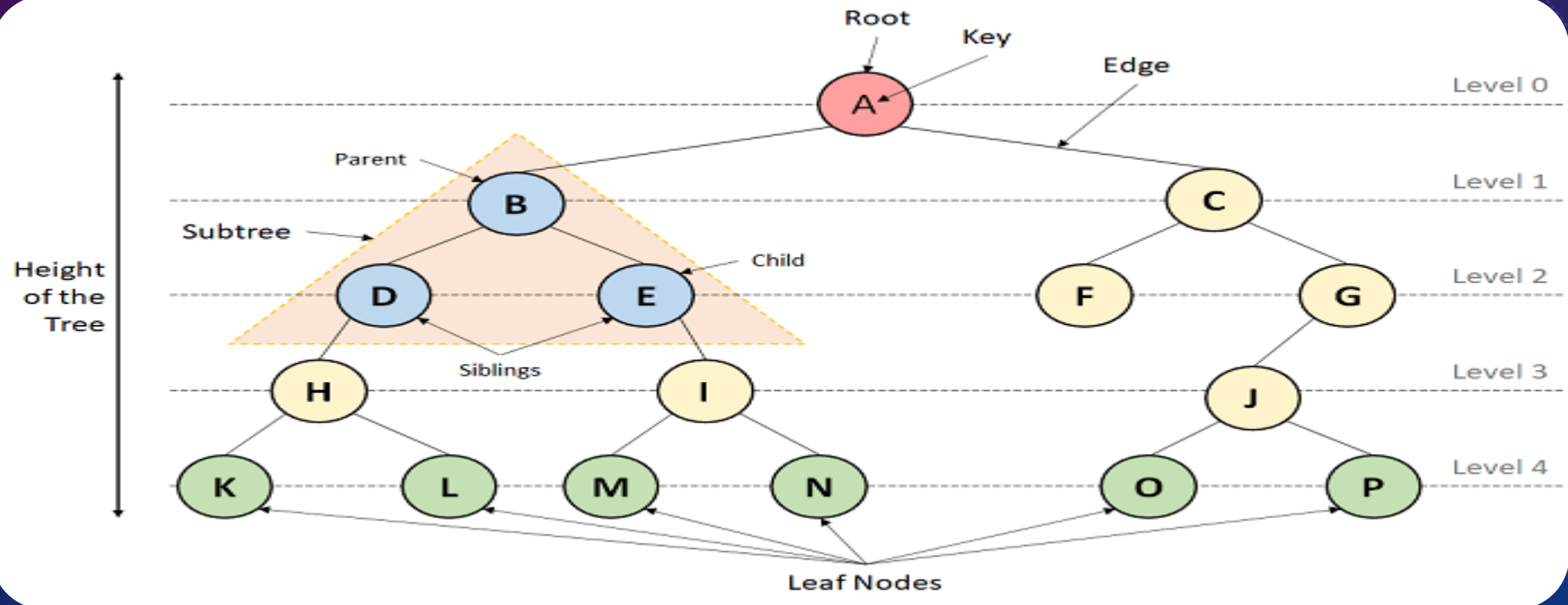


NON-LINEAR DATA STRUCTURE – TREES

Similar to a graph, a tree is also a collection of vertices and edges.

However, in tree data structure, there can only be one edge between two vertices.

It has a root node (parent) and leaf nodes (children) that forms a hierarchical tree structure



THAT'S ALL
